Fake News Detection

SUFFICIENT LIBRARIES

In [ ]:
```
import pandas as pd
import nltk
from sklearn.preprocessing import LabelEncoder
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from sklearn.feature_extraction.text import TfidfVectorizer
```

Load Data Set

In [ ]:
```
import pandas as pd
DF = pd.read_csv("/content/drive/MyDrive/Data_Science_Projects/combined_fake_r
```

Inspect Data_set

In [ ]:
```
DF.head()
```

Out[ ]:

	source	text	label	title	url
0	BBC News	Highlights from Grace Road in Leicester as Eng...	1	Women's ODI Cricket	https://www.bbc.co.uk/iplayer/episode/m002d4yf...
1	CNN	For months, President Donald Trump has suggest...	1	Why Trump has struggled to get Xi on the phone...	https://www.cnn.com/2025/06/04/politics/trump-...
2	CNN	A group of migrants the Trump administration s...	1	Migrants flown to El Salvador under Alien Enem...	https://www.cnn.com/2025/06/04/politics/migran...
3	CNN	The family of a teenager from Massachusetts wh...	1	‘Please, get him out’: Little brother of detai...	https://www.cnn.com/2025/06/04/us/video/teen-d...
4	CNN	Hackers are tricking employees at companies in...	1	Hackers are using a modified Salesforce app to...	https://www.cnn.com/2025/06/04/tech/hackers-ab...

In [ ]:
```
DF.tail()
```

Out[ ]:

	source	text	label	title	
80766	The New York Times	PORT TRAVIS, BRITISH INDIAN OCEAN TERRITORY (C...	0	Russian President Implements Revolutionary Cli...	https://www.thenewyorktimes.c world/2025/0
80767	NPR	NORTH ERIKA, ALGERIA (March 13, 2026) - In a g...	0	Japanese Prime Minister Implements Revolutiona...	https://www.npr.com/world/20 05/30/japanes
80768	Fox News	WEST CHRISTINA, UKRAINE (March 28, 2026) - In ...	0	European Union Reveals Surprising Cybersecurit...	https://www.foxnews.com/wo 2026/05/15/euro
80769	Al Jazeera	NORTH GREGORYBURY, CYPRUS (August 09, 2025) - ...	0	International Monetary Fund Declares Major Shi...	https://www.aljazeera.com/wo 2025/06/14/i
80770	The New York Times	EAST KATHRYNMOUTH, ZAMBIA (June 03, 2026) - In...	0	Apple Announces Breakthrough in Technology Usi...	https://www.thenewyorktimes.c world/2026/0

In [ ]:
```
DF.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 80771 entries, 0 to 80770
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   source  80771 non-null  object
 1   text    80770 non-null  object
 2   label   80771 non-null  int64 
 3   title   80771 non-null  object
 4   url     29042 non-null  object
dtypes: int64(1), object(4)
memory usage: 3.1+ MB
```

In [ ]:
```
DF.describe()
```

Out[ ]:

	label
count	80771.000000
mean	0.586423
std	0.492477
min	0.000000
25%	0.000000
50%	1.000000
75%	1.000000
max	1.000000

Preprocessing

Checking Duplicates

In [ ]:
```
DF.duplicated().sum()
```

Out[ ]:
```
np.int64(0)
```

Checking Irrelevant Features

In [ ]:
```
DF.columns
```

Out[ ]:
```
Index(['source', 'text', 'label', 'title', 'url'], dtype='object')
```

In [ ]:
```
DF.drop("url",axis =1,inplace = True)
```

In [ ]:
```
DF.columns
```

Out[ ]: Index(['source', 'text', 'label', 'title'], dtype='object')

Checking Null Values

In [ ]:
```
DF.isna().sum()
```

Out[ ]:

	0
source	0
text	1
label	0
title	0

dtype: int64

In [ ]:
```
DF[DF["text"].isna()].index
```

Out[ ]: Index([41705], dtype='int64')

In [ ]:
```
DF.drop(41705, inplace = True , )
DF.reset_index(drop=True, inplace=True)
```

In [ ]:
```
DF.isna().sum()
```

Out[ ]:

	0
source	0
text	0
label	0
title	0

dtype: int64

Checking Data Type

In [ ]:
```
DF.dtypes
```

Out[ ]:

	0
source	object
text	object
label	int64
title	object

dtype: object

In [ ]:
```
from sklearn.preprocessing import LabelEncoder
LB = LabelEncoder()
DF.source = LB.fit_transform(DF["source"])
```

In [ ]:
```
DF.source.unique()
```

Out[ ]:
```
array([ 4,  5,  2, 53,  0, 14, 60, 50, 48, 17, 51, 61, 55, 56, 52,  3,  8,
       54, 58, 57, 59, 45, 32, 49, 27, 39, 16, 23, 34, 46, 40, 35, 24, 44,
       31, 30, 22, 13, 33, 37, 25, 20, 18, 47, 19, 42, 28, 29, 21, 26, 41,
       43, 36, 15, 38,  6, 10, 12,  7,  9,  1, 11])
```

In [ ]:
```
DF.label.unique()
```

Out[ ]:
```
array([1, 0])
```

Split Feature's & Target

In [ ]:
```
tem = DF["source"].astype(str) + " " + DF["text"] + " " + DF["title"]
```

In [ ]:
```
tem[1]
```

Out[ ]:
```
'5 For months, President Donald Trump has suggested his personal relationship
with Chinas Xi Jinping will be what unlocks the worlds most consequential tra
de dispute.\r\nYet he couldnt get Xi on the telep… [+6200 chars] Why Trump ha
s struggled to get Xi on the phone to talk trade | CNN Politics'
```

In [ ]:
```
x= DF["source"].astype(str) + " " + DF["text"] + " " + DF["title"]
y = DF.label
```

Tokenization

In [ ]:
```
from nltk import TweetTokenizer
TT = TweetTokenizer()
```

In [ ]:
```
x = x.apply(TT.tokenize)
```

Removing StopWords

In [ ]:
```
import nltk
from nltk.corpus import stopwords
nltk.download("stopwords")
SW = stopwords.words("english")
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

In [ ]:
```
x = x.apply(lambda line : [ i.lower() for i in line if i.lower() not in SW])
```

Stemming OR Lemmatizaton : Here we go with Lemmatizaton

In [ ]:
```
from nltk.stem import WordNetLemmatizer
nltk.download("wordnet")
WNL = WordNetLemmatizer()
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
```

In [ ]:
```
x = x.apply(lambda line : [WNL.lemmatize(i) for i in line])
```

Rejoin

In [ ]:
```
x = x.apply(lambda line : " ".join(line))
```

Ngram & Vectorization

In [ ]:
```
from sklearn.feature_extraction.text import TfidfVectorizer
vec = TfidfVectorizer(ngram_range= (1,3))
```

In [ ]:
```
x= vec.fit_transform(x)
```

Split into Training and Testing Sets

In [ ]:
```
from sklearn.model_selection import train_test_split
```

In [ ]:
```
x_train , x_test , y_train, y_test  = train_test_split(x,y,test_size = 0.2,ran
```

Model Creation

In [ ]:
```
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier()
```

In [ ]:
```
knn.fit(x_train,y_train)
```

Out[ ]:
```
▾ KNeighborsClassifier
KNeighborsClassifier()
```

In [ ]:
```
y_pred = knn.predict(x_test)
```

Checking Perfomance Score

In [ ]:
```
from sklearn.metrics import classification_report
print(classification_report(y_pred,y_test))
```

```
              precision    recall  f1-score   support

           0       0.71      0.80      0.75      6031
           1       0.87      0.80      0.83     10123

    accuracy                           0.80     16154
   macro avg       0.79      0.80      0.79     16154
weighted avg       0.81      0.80      0.80     16154
```

now we got 80% accuracy on this model